

Literature-Implied Negative Answer to Problem 2.13 of arXiv:0710.5396

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11 August 2026

Status

`literature_implied_answer`; full negative answer. Both questions in Problem 2.13 of Richard J. Smith, *Gruenhage compacta and strictly convex dual norms*, arXiv:0710.5396, are false.

Source question

On PDF page 9, at the end of Section 2, Problem 2.13 asks:

If $C(K)^*$ admits a strictly convex dual norm then is K Gruenhage? More ambitiously, if X^* is a dual Banach space with strictly convex dual norm, is B_{X^*} Gruenhage?

The topology on a dual ball here is the weak-star topology.

Decisive later result

Richard J. Smith, *Tree Duplicates, G_δ -diagonals and Gruenhage spaces*, arXiv:1102.0982, gives the needed ZFC counterexample. Its Theorem 1.4, quoted from Theorem 3.1 of Orihuela–Smith–Troyanski, arXiv:1012.5595, says that for scattered compact K , property $(*)$ is equivalent to the existence on $C(K)$ of an equivalent norm with strictly convex dual norm. Corollary 2.5 then states that the one-point compactification K of the constructed space D is scattered, compact, non-Gruenhage, and has $(*)$; consequently $C(K)$ admits an equivalent lattice norm with strictly convex dual norm.

Thus $C(K)^*$ has a strictly convex dual norm although K is not Gruenhage, which answers the first question negatively.

Why the same example answers the ambitious question

Let $X = C(K)$ with the equivalent predual norm from Corollary 2.5, and write $\|\cdot\|$ for its strictly convex dual norm. Equivalence with the canonical dual norm gives

$$M := \sup_{x \in K} \|\delta_x\| < \infty.$$

Choose a fixed $a > M$. The map

$$K \longrightarrow B_{X^*}, \quad x \longmapsto a^{-1}\delta_x$$

is a weak-star homeomorphic embedding: each coordinate against $f \in C(K)$ is $a^{-1}f(x)$, and compactness of K makes the continuous injection a homeomorphism onto its image. Gruenhage spaces are hereditary, since one may restrict every witnessing open family to a subspace. If (B_{X^*}, w^*) were Gruenhage, its embedded copy of K would be Gruenhage, contradicting Corollary 2.5. Hence the ambitious dual-ball assertion is false as well.

Provenance and scope

The later paper cites Smith's 2009 article and states the exact $C(K)$ counterexample, but does not label Corollary 2.5 as an answer to Problem 2.13. The first implication is immediate from that corollary; the second uses the fixed-scalar Dirac embedding above. This is therefore a literature-implied answer rather than a newly discovered counterexample. Both parts of the source problem are settled; no stated subcase remains open.

References

- [1] R. J. Smith, *Gruenhage compacta and strictly convex dual norms*, J. Math. Anal. Appl. 350 (2009), 745–757; arXiv:0710.5396.
- [2] J. Orihuela, R. J. Smith, and S. Troyanski, *Strictly convex norms and topology*, Proc. London Math. Soc. 104 (2012), 197–222; arXiv:1012.5595.
- [3] R. J. Smith, *Tree Duplicates, G_δ -diagonals and Gruenhage spaces*, Topology Appl. 158 (2011), 1813–1818; arXiv:1102.0982.